

Imaging techniques (ultrasound)

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Introduction

The health status of soft tissue can be estimated by measuring its elasticity, which has an impact on tissue motion. The examination of soft tissue with conventional ultrasound helps to detect lesions in the fascia, muscles and organs, but without gaining information about its elastic properties. The measurement of elastic properties can be achieved by tracking the deformation of soft tissue in B-Mode images using speckle tracking or by using cross-correlation techniques, either as direct motion measurement or elastography for strain assessment. Finding the areas of altered elasticity and reinstating normal elasticity through appropriate treatment makes tissue movement analysis and elastography an important diagnostic feature for treatment decisions and for monitoring the reaction to different treatment modalities.

In vivo evaluation of connective tissue movement

Sonography is the only method that allows the clinician a radiation-free evaluation of tissue movement with the necessary temporal and graphical resolution *in vivo*. When recording high-definition ultrasound videos, these can be used to track the movement of connective tissue. One possibility is the cross-correlation method developed in MATLAB (MathWorks, Natick, MA) by Dilley et al. (2001). It is basically a speckle- or pixel-tracking algorithm that uses the digital information of video data to track the movement

of selected regions. The method is described and discussed in the following section.

Cross-correlation analysis

As previously mentioned, the cross-correlation analysis uses videos recorded with a high-definition ultrasound device to track the movement of selected areas or tissues. In the first step, the video is loaded into the software and converted into individual gray level frames. In the next step, you can select as many rectangle-shaped regions of interest (ROIs) as you wish to track from the first frame of the video. In the actual analysis, the correlation coefficient between the pixel gray levels for all selected ROIs in two adjacent images is calculated. The pixel shift providing the maximum correlation coefficient corresponds to the relative movement between two frames (Dilley et al., 2001). This is repeated for all adjacent frames to calculate the movement of the ROIs during the recorded period. To control background movement, an additional ROI can be placed on the skin or a bony structure, which remain static during the movement that is being tracked. The background movement is then subtracted from the original results.

The method was originally developed for and has been extensively used to quantify nerve movement *in vivo*. Studies on reliability show promising results, so it has been described as a reliable method with an ICC ranging from 0.70–0.99 (Boyd et al., 2012; Carroll et al., 2012; Dilley et al., 2001, 2007; Ellis et al., 2008, 2012).

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It has also been used in research to quantify the movement of connective tissue during passive movement. Langevin et al. (2011) were the first to show changes in fascial shear strain in the thoracolumbar fascia in patients with unspecific low back. More recently, Griefahn et al. (2017) used a similar approach to quantify the fascial mobility of the thoracolumbar fascia after foam rolling at the dorsal body. Unfortunately, no measures for reliability were calculated in these studies.

In our laboratory, we used the cross-correlation method to quantify the movement of different

layers of the fascia lata at the anterior thigh. Here, movement of the superficial ($ICC_{(3,1)} = 0.88$ (0.75–0.95), $P < 0.0001$) and deep layer ($ICC_{(3,1)} = 0.77$ (0.55–0.91), $P < 0.0001$) showed good reliability (Krause, 2018). To provide a better understanding of the methodology of the cross-correlation, our research approach is described below.

To evaluate the movement of different fascial layers at the anterior thigh, six equidistant ROIs per fascial layer were chosen to steadily evaluate movement of the layers. Figure 20.1 shows the original US signal at the beginning of the video (Figure 20.1A) and the six ROIs per fascial layer

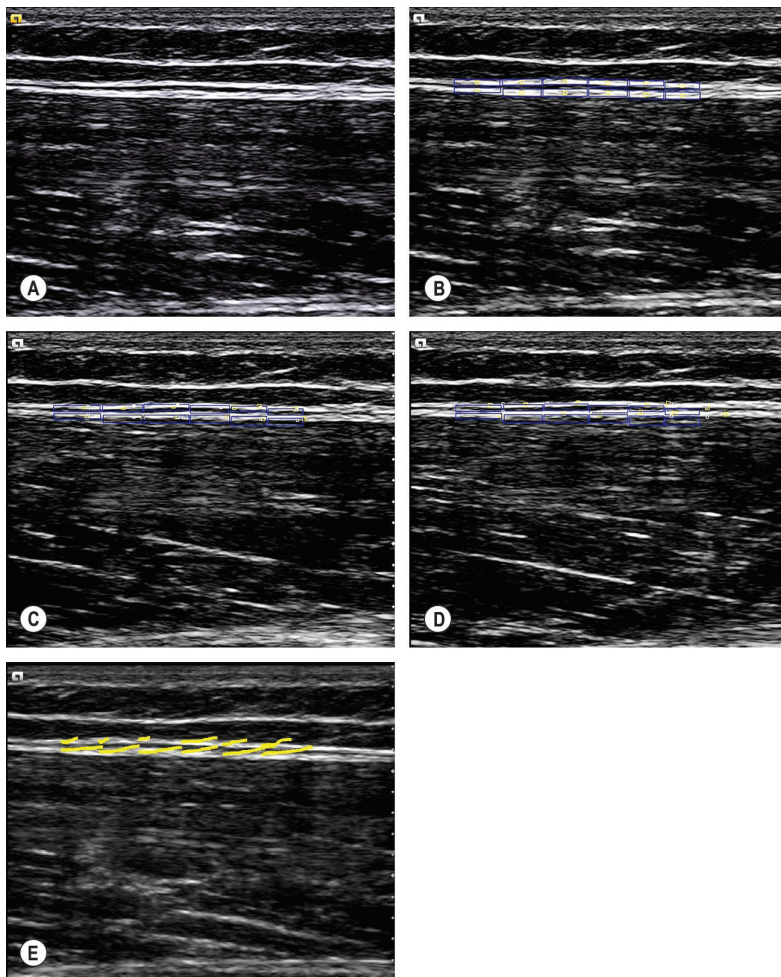


Figure 20.1

20.1A Raw ultrasound video signal at the beginning of the recorded video.

Structures (top to bottom): skin, subcutaneous fat, subcutaneous connective tissue, superficial layer of deep fascia, loose connective tissue (enriched with hyaluronic acid), deep layer of the deep fascia and rectus femoris muscle.

20.1B Selection of six equidistant ROIs in each fascial layer in the first video frame.

20.1C Graphical results of the cross-correlation analysis after 5 s of the recorded video.

20.1D Graphical results of the cross-correlation analysis after 10 s of the recorded video.

20.1E Resulting line graph of the movement of all selected ROIs.

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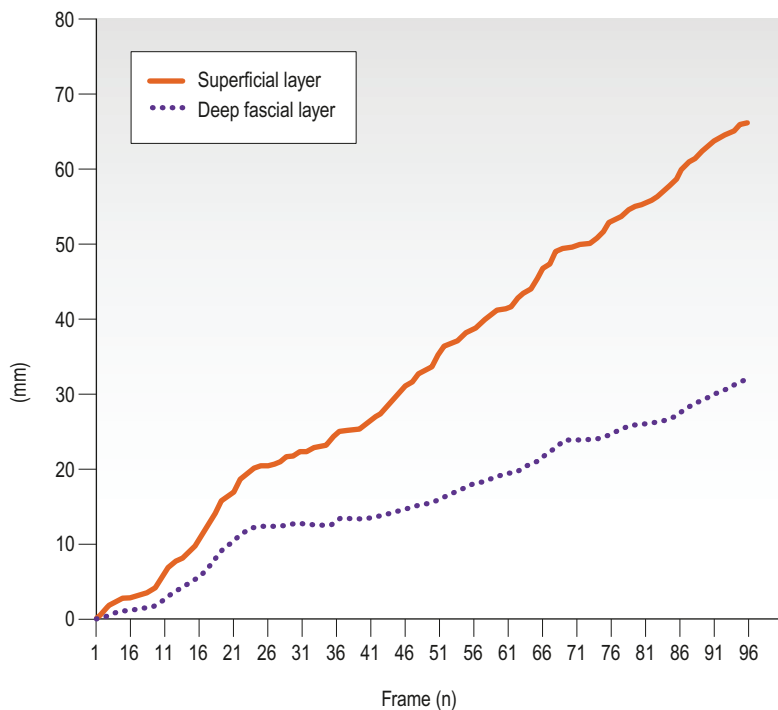


Figure 20.2

Movement of the superficial and the deep fascial layer during the recorded video sequence.

(Figure 20.1B). The movement of each individual ROI in the middle (5 s) and at the end (10 s) of the recorded video is shown in Figures 20.1C and 20.1D, respectively. Figure 20.1E shows the resulting line graph of the movement of all selected ROIs.

The blue boxes are the selected ROIs in each fascial layer, the small white squares represent the mean movement of all twelve ROIs, while the slightly bigger yellow squares represent the movement of individual ROIs. To quantify the movement of each layer, the mean movement of the six ROIs per fascial layer was calculated (Figure 20.2).

Traditionally, ultrasound has extensively been used in internal medicine, cardiology and gynecology. However, new improvements in temporal

and graphical resolution have led to increased use in orthopedics as well as in exercise and sport science research in recent years. The cross-correlation analysis is a useful method to extract quantitative information on tissue movement from high-frequency ultrasound data.

Ultrasound elastography

Introduction

Manual palpation is an important assessment tool for estimating the pathological state of soft tissue. The goal is to identify myofascial areas that are structurally or functionally compromised. Surface pressure application gives feedback about the tissue strain, and skin drag provides information about shear strain. Communication about these findings is difficult

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because of a lack of numerical values that would accurately quantify them. Palpation is often limited to larger superficial structures with a high stiffness contrast compared to the surrounding areas. Smaller, areas with lower stiffness contrast or deep lying changes can often be overlooked even though they can play an important role in pain and dysfunction.

Conditions causing myofascial stiffness

Soft tissue health can be reflected by the degree of stiffness changes caused by various conditions and in sports often through neurogenic inflammation (NI) as a reaction to trauma of any kind (Toumi and Best, 2003); conversely tendons can react with a softening (Aubry et al., 2015). With micro-analytical techniques *in vivo*, it is possible to identify those biochemicals which are associated with NI of the myofascia in active myofascial trigger points (Shah et al., 2008). Under experimental conditions increase of fascia stiffness can be related to isometric strain and changes in tissue hydration (Schleip et al., 2012). Clinically myofascial trigger points, fibrosis and or densification are possible causes of increased fascial stiffness.

Myofascial stiffness can affect the joint range of motion, fascial glide, strength, posture and motor unit recruitment measured with electromyography. Conventional X-ray, MRI, CT-scan or ultrasound cannot identify stiffness, but magnetic resonance elastography (MRE) and forms of ultrasound elastography (UE) can.

Ultrasound elastography methods

Besides MRE which is exclusively available for research, strain elastography (SE) and shear wave elastography (SWE) are in clinical use for tumor detection, staging of liver fibrosis and musculoskeletal applications. Both methods

differ by the way of stress application to create tissue-motion, by their results—quantitative vs. qualitative—operator dependency, depth of penetration and the size of the region of interest (ROI).

Strain elastography

SE (synonym: compression-, or static-elastography) was first introduced by Ophir et al. (1991). It requires rhythmic manual pressure on the surface of the tissue with the ultrasound transducer producing a compressive strain. Compressive strain is the ratio of an objects change in thickness Δl to its initial thickness l (Compressive strain = $\Delta l/l$) which is dimensionless (Figure 20.3).

An elastogram is an image with color-coded strain representation. In stiff tissue the strain is small; in softer tissue high (Garra, 2007). SE produces qualitative images and requires areas of different strain within the region of interest (ROI). The consequence is that equally stiff tissue within the ROI does not give a result because of a lack of strain differences. From the qualitative elastogram numeric values can be obtained by counting the pixels which represent hard or soft with the software Image J (Abramoff et al., 2004). SE allows semi-quantitative measurements by calculating a strain ratio (SR) of two different areas. It is calculated by dividing the strain of a reference area by the strain of a target area. $SR = \text{Strain of reference (B)} / \text{Strain of a target (A)}$. When the target area has a higher strain (softer) the SR is >1 , when the target area has a lower strain (stiffer) the SR is <1 . A Strain elastogram (Figure 20.4) shows a strain ratio of 2.59 between the fascia (Target 1A) with a strain of 0.19% and the muscle (Reference 2B) 0.50%. The fascia has a lower strain value, which means it is 2.59 times stiffer compared to the muscle with higher strain.

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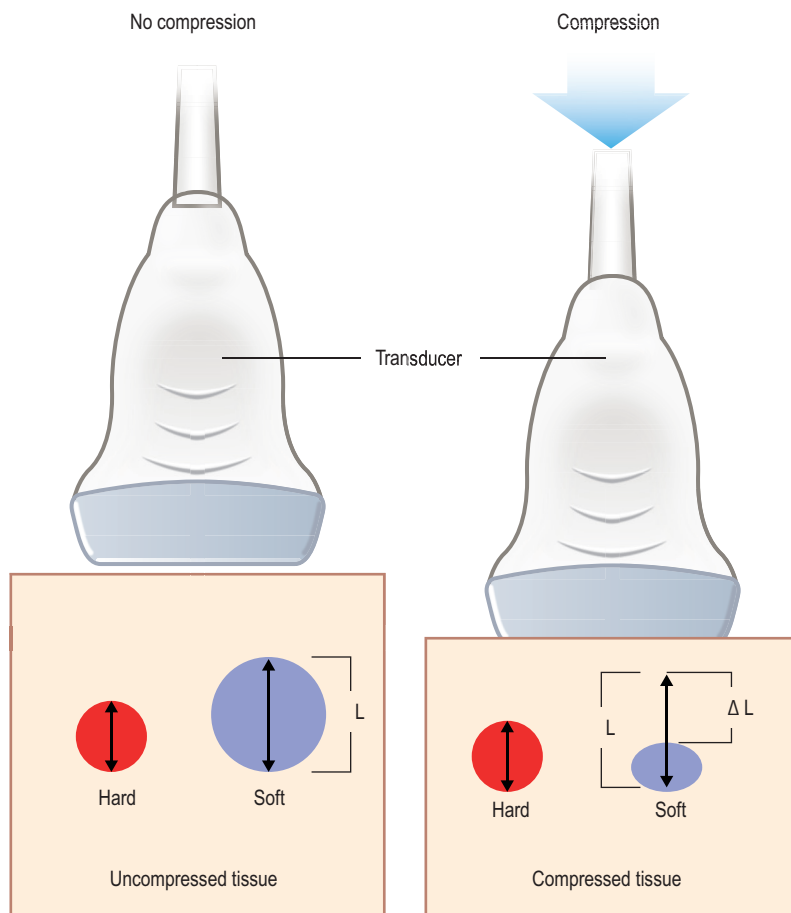


Figure 20.3

Through a rhythmic compression of the tissue a time shift occurs in the echo signal which is proportional to the strain of the tissue.

The quality of the elastogram can be strongly operator dependent when the compression/decompression cycle requires an amplitude of several millimeters. Newer technologies need minimal transducer motion for excitation or even use internal motion generated through pulse or breathing to obtain the elastogram. Interrater reliability studies show a relatively good agreement provided the tissue is not too soft (Boettner 2018). When the transducer is driven by a computer-controlled shaker, all operator induced artefacts can be eliminated resulting in a stable reproducible elastogram (Bauermeister 2017).

Shear wave elastography

Shear wave elastography was first mentioned in 1995 (Sarvazyan et al. 1995) as a quantitative measure of tissue stiffness versus the qualitative assessment with SE. Instead of manual compression in SE with a tissue displacement up to 3 mm like, tissue peak displacement in the order of i.e. 10 μm or more is achieved through a push beam generated with an acoustic radiation force impulse (ARFI) emitted from the ultrasound probe (Nightingale et al., 2002). The resulting shear waves are perpendicular to the direction of excitation force in a transverse

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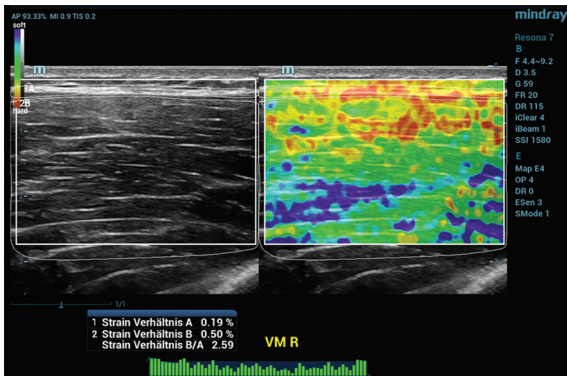


Figure 20.4

Mindray Resona 7 Strain elastography. The fascia (Target 1A) with a strain of 0.19% and the muscle (Reference 2B) 0.50%. The fascia has a lower strain value, which means it is 2.59 times stiffer compared to the muscle with higher strain.

direction (Figure 20.5). The resulting shear waves detected through the tracking beam of the ultrasound probe which generates the push beam. In Figure 20.6 the shear waves are shown as they travel through the ROI. Parallel lines indicate reliable, distorted lines unreliable measurements.

The push beam has a duration of less than a microsecond, the resulting shear waves persist for several milliseconds (Sarvazyan et. al. 1998). By measuring the peak to peak duration, the speed of the shear wave v (nu) propagation expressed in m/sec can be derived. The propagation speed v is higher in stiff and lower in soft tissue. From the propagation speed v the Shear Modulus G in kPa or Young's Modulus E in kPa can be estimated with the formula: $E = 3G = 3\rho v^2$. In this formula ρ (rho) is the mass density of the tissue, assumed to be ≈ 1 . Figure 20.7 shows a shear wave elastogram of the thoraco lumbar fascia TLF (Trace 1) with a mean Young's

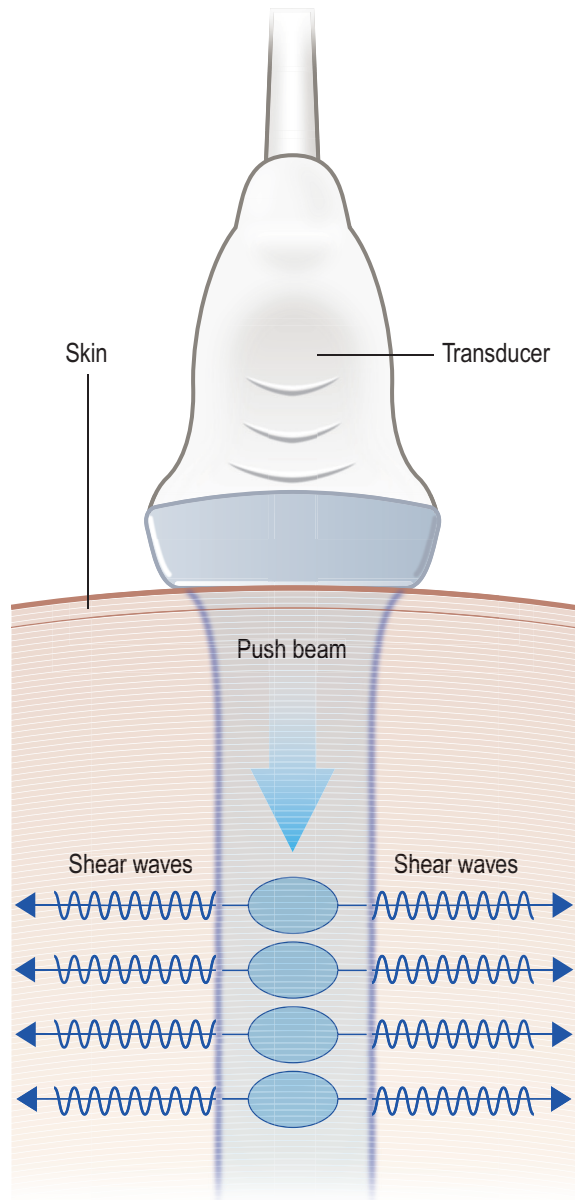


Figure 20.5

The resulting shear waves are perpendicular to the direction of the push beam in a transverse direction.

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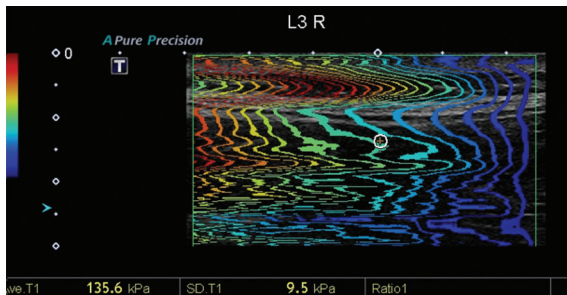


Figure 20.6

Canon Aplio 500. Shear wave pattern. Parallel lines indicate reliable—distorted lines unreliable—measurements.

modulus of 226.01 kPa and muscle (Trace 2) with a mean Young's modulus of 40.12 kPa.

Other shear wave forms are supersonic shear wave imaging (SSI) (Bercoff et.al., 2004) which has been used widely in muscle research and acoustic radiation force impulse imaging (ARFI) (Sarvazyan et.al., 2011).

Ultrasound elastography research of the myofascial and clinical application

The focus of myofascial research with ultrasound-elastography is to establish normative values for muscle, fascia and tendons. In sports medicine injury prevention is a major concern where early recognition of increased myofascial stiffness could play an important role. A pilot study on 20 juvenile soccer players, mean age 15 years, showed a statistical difference for the rectus femoris muscle/fascia and the fascia of the vastus medialis and adductor magnus (highlighted yellow). (Table 20.1) (K. Bauermeister, 2021). Table 20.2 shows measurements of muscles, the thoraco lumbar and the plantar fascia of 45 males, mean age 35 years, with a high physical work load with statistically significant left/right differences of the

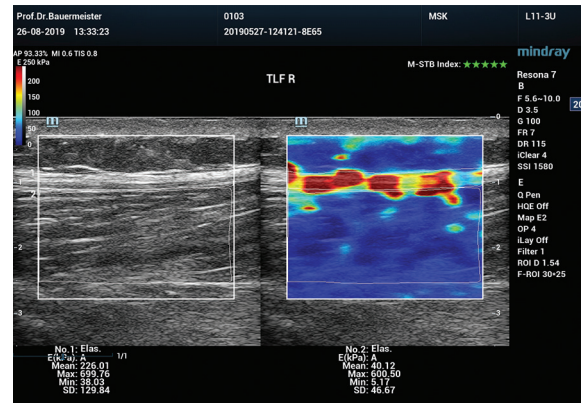


Figure 20.7

Mindray Resona 7. Shear wave elastogram of the thoraco lumbar fascia TLF (Trace 1) with a mean Young's modulus of 226.01 kPa and muscle (Trace 2) with a mean Young's modulus of 40.12 kPa

rectus femoris and the plantar fascia (highlighted yellow). Table 20.3 is an example of how to translate SWE measurements into a treatment strategy. Here the patient is its own control by comparing the left to the right side. Significant discrepancies between the left and the right side are marked in red. The effect of a therapeutic intervention can be monitored by reevaluating at the same measurement points after the intervention.

Comparing SWE results obtained with equipment from different manufacturers can produce different results. There are no agreed standards about the generation and frequency of the push beam. The size of the area where measurements within the ROI can be made varies widely.

They range from some square millimeters to several square centimeters. Comparing study results is difficult because researches arbitrarily chose regions of the fascia or within the muscles which appear homogeneous to them. Areas with very high kPa-values are usually omitted

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Table 20.1 Medians and means of Young's moduli in left/right comparison. Significant differences are highlighted yellow (see text page 231)

Region	Fascia left	Fascia right	Muscle left	Muscle right
Tensor fasciae latae	43.33 (17.75)	43.85 (20.06)	30.82 (13.65)	32.03 (9.63)
Rectus femoris middle	24.19 (8.91)	20.46 (9.02)	20.62 ± 8.34	17.36 ± 3.56
Distal rectus femoris	22.46 (20.35)	22.91 (14.12)	26.46 (38.98)	29.51 (22.73)
Vastus medialis	13.73 (4.70)	11.53 (5.60)	13.67 (3.04)	13.43 (3.60)
Vastus lateralis	111.52 ± 37.12	115.84 ± 37.04	28.65 (11.03)	30.34 (9.52)
Tibialis anterior	91.28 (34.2)	34.63 (6.41)	34.86 ± 5.21	34.63 ± 6.41
Peroneus longus	82.64 ± 27.23	78.56 ± 21.83	27.12 (8.57)	27.94 (8.57)
Gluteus medius	23.10 (38.73)	17.46 (27.66)	21.75 ± 7.81	19.95 ± 7.54
Gluteus maximus	9.38 (2.33)	10.28 (2.88)	20.33 ± 4.75	20.31 ± 5.60
Adductor magnus	13.98 (4.73)	15.86 (5.73)	15.79 (3.21)	17.24 (4.12)
Biceps femoris	31.99(14.39)	32.57 (19.35)	18.41 (4.59)	19.35 (4.44)
Gastrocnemius medial head	33.70 (16.04)	38.27 (22.14)	20.42 ± 6.24	19.52 ± 4.11
Gastrocnemius lateral head	42.22 ± 17.55	43.91 ± 14.85	18.92 ± 4.34	19.31 ± 4.91

Table 20.2 Medians and means of Young's moduli in left/right comparison. Significant differences are highlighted yellow (see text page 231)

Region	Left	Right
Adductor magnus	13.91 (6.59)	15.1 (7.63)
Biceps femoris middle	12.26 (3.95)	12.64 (4.95)
Erector spinae L4	25.57 ± 4.75	25.8 ± 6.12
Gastrocnemius lateral head	14.00 (4.50)	13.00 (5.25)
Gluteus maximus	9.17 (5.62)	9.17 (3.09)
Gluteus medius	9.00 (3.00)	8.00 (3.00)
Plantar fascia	27.51 ± 8.67	23.99 ± 6.65
Rectus femoris middle	15.50 (6.00)	14.00 (5.00)
Tibialis anterior	36.00 (7.50)	36.00 (9.00)
Thoraco lumbar fascia TLF L4	58.86 ± 30.18	55.57 ± 29.54
Upper trapezius	25.00 (17.00)	26.00 (11.00)

because most researchers don't know how to interpret these "red and brown dots" within them. Especially these areas can provide valuable information for practitioners in sports- and pain medicine. They can represent "disease" state like MTrPs with increased neurogenic inflammation which develop through overuse or trauma.

SE has its place in research and clinical use as well even though it does not produce quantitative data like SWE. It can provide valuable information when the elastograms are analyzed for the number of hard pixels comparing the myofascia of the left and the right side. The validity of these findings has been shown in several preliminary studies. The stiffness patterns are correlated to pain location, range of motion (Bauermeister, 2012), pressure pain threshold and tissue indentometry (Bauermeister, 2015; Farasyn and Lassat, 2015).

Conclusion

Soft tissue pain and dysfunction is a problem of altered tissue stiffness not just in one myofascial location but along myofascial chains requiring an assessment of the entire myofascia from head to toe. SE is a qualitative and semi quantitative method which is integrated even into small portable ultrasound devices. It is useful in clinical practice for identifying the stiffest myofascial areas in all tissue layers. It serves as an objective enhancement of manual palpation.

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Table 20.3 Young's moduli in left right comparison in one pain patient with mean, maximum and minimum values. Significant differences between both sides are highlighted in red [see text page 231].

Region	E Mean L	E Mean R	E Max L	E Max R	E Min L	E Min R
Upper trapezius upright	42.71	54.93	172.08	228.04	4.84	4.17
Upper trapezius supine	53.30	32.06	189.80	204.63	5.16	0.17
Thoracic 1	31.99	37.62	267.38	340.05	4.01	5.45
Thoracic 2	37.55	33.07	467.69	357.50	3.36	3.60
Thoracic 3	25.97	24.48	74.89	174.48	0.13	1.34
Thoracic 4	25.73	17.72	126.78	83.55	6.15	0.02
Thoracic 5	24.88	19.36	113.96	226.67	4.12	4.48
Lumbar 1	22.29	16.96	76.42	83.90	4.43	2.35
Lumbar 2	28.59	16.74	125.96	73.93	4.31	3.86
Lumbar 3	25.39	20.35	141.70	89.65	0.99	1.31
Lumbar 4	27.27	20.09	154.60	167.16	3.64	2.96
Sacrum	24.24	25.92	208.97	168.89	0.48	0.00
Gluteus medius	25.46	12.68	210.19	110.74	3.01	0.00
Biceps femoris middle	14.53	15.47	94.39	67.77	1.74	3.17
Biceps femoris distal	27.61	17.66	185.55	99.97	2.70	0.28
Biceps femoris proximal	11.17	14.85	63.13	66.91	2.84	3.02
Gastrocnemius lateral	32.25	17.94	201.88	105.12	1.48	0.01
Gastrocnemius medial	22.90	19.25	161.15	234.21	1.74	0.99
Adductor magnus	17.53	11.01	84.88	55.56	3.68	2.91
Soleus	20.84	28.57	295.90	232.92	3.22	4.53
Plantar fascia	25.30	26.79	75.70	59.00	2.15	6.56
Rectus femoris middle	34.76	22.77	323.70	77.14	4.39	6.72
Tibialis anterior	38.41	39.04	230.84	147.17	14.94	14.48
Tensor fasciae latae	37.55	25.56	124.16	127.69	5.66	2.96

SWE is a quantitative method which measures the shear wave speed. To obtain individual values of shear wave speed, one or several areas within the ROI must be chosen for analysis. Some devices offer immediate results of shear wave speed or Young's modulus over the entire ROI with mean, maximum, minimum and standard deviation

values. More research is needed to obtain representative standard values for all relevant areas of the myofascia. As long as we do not have normative values we need to use the patient as their own control. This is at present the best approach to decide on a target area for treatment.

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References

- Abramoff, M.D., Magelhaes, P., Ram, S.J. (2004) Image processing with image. *J Bio-photonics Int.* 11: 36–42.
- Aubry, S., Nueffer, J.P., Tanter, M., Becce, F., Vidal, C. & Michel, F. (2015) Viscoelasticity in Achilles tendonopathy: quantitative assessment by using real-time shear-wave elastography. *Radiology*. 274: 821–829.
- Bauermeister, W. (2012) Ultraschall-Elastographie zur Diagnose myofaszi-aler Schmerzsyndrome. Der Deutsche Schmerz- und Palliativtag 2012. Frankfurt, Germany.
- Bauermeister, W. (2015) Ultrasound elastography for the evaluation of the elastic properties of fascia and muscle. Fourth International Fascia Research Congress, 2015. Reston, VA, USA.
- Bauermeister, W. (2017) Optimierung des Strain-Elastografie durch Computer assistierte Steuerung der Ultraschall-sonde. *Ultraschall in Med.* 38: P2.011.
- Bauermeister, K. (2021) The role of elastography in sports injury prevention. School of Medicine, Technical University Munich, unpublished.
- Böttner, C. (2018) Interrater reliability of ultrasound elastography to examine the stiffness and elasticity of muscles and fascia of the lower extremity in healthy adults. *J Bodyw Mov Ther.* 22(4): 849–850. doi:10.1016/j.jbmt.2018.09.015
- Boyd, B.S., Gray, A.T., Dilley, A., et al. (2012) The pattern of tibial nerve excursion with active ankle dorsiflexion is different in older people with diabetes mellitus. *Clin Biomech.* 27: 967–971.
- Carroll, M., Yau, J., Rome, K., et al. (2012) Measurement of tibial nerve excursion during ankle joint dorsiflexion in a weight-bearing position with ultrasound imaging. *J Foot Ankle Res.* 5: 1.
- Dilley, A., Greening, J., Lynn, B., et al. (2001) The use of cross-correlation analysis between high-frequency ultrasound images to measure longitudinal median nerve movement. *Ultrasound Med Biol.* 27: 1211–1218.
- Dilley, A., Summerhayes, C. & Lynn, B. (2007) An in vivo investigation of ulnar nerve sliding during upper limb movements. *Clin Biomech (Bristol, Avon).* 22(7): 774–779.
- Ellis, R., Hing, W., Dilley, A., et al. (2008) Reliability of measuring sciatic and tibial nerve movement with diagnostic ultrasound during a neural mobilisation technique. *Ultrasound Med Biol.* 34: 1209–1216.
- Ellis, R.F., Hing, W.A. & McNair, P.J. (2012) Comparison of longitudinal sciatic nerve movement with different mobilization exercises: an in vivo study utilizing ultrasound imaging. *J Orthop Sports Phys Ther.* 42: 667–675.
- Farasyn, A. & Lassat, B. (2015) Cross friction algometry (CFA): comparison of pressure pain thresholds between patients with chronic non-specific low back pain and healthy subjects. *J Bodyw Mov Ther.* 20: 224–234.
- Garra, B.S. (2007) Imaging and estimation of tissue elasticity by ultrasound. *Ultrasound Q.* 23: 255–268.
- Griefahn, A., Oehlmann, J., Zalpour, C., et al. (2017) Do exercises with the foam roller have a short-term impact on the thoracolumbar fascia? A randomized controlled trial. *J Bodyw Mov Ther.* 21: 186–193.
- Krause, F. (2018) *Akuteffekte von Self-Myofascial-Release auf Beweglichkeit, passive Gewebesteifigkeit, Dehnwahrnehmung und faszi-ale Gleitbewegung—Eine randomisierte & kontrollierte Cross-Over Studie.* Inaugural dissertation. Frankfurt am Main, Germany.
- Langevin, H.M., Fox, J.R., Koptiuch, C., et al. (2011) Reduced thoracolumbar fascia shear strain in human chronic low back pain. *BMC Musculoskeletal Disord.* 12: 203.
- Ophir, J., Cespedes, I., Ponnekanti, H., Yazdi, Y. & Li, X. (1991) Elastography: a quantitative method for imaging the elasticity of biological tissues. *Ultrason Imaging.* 13: 111–134.
- Sarvazyan, A., Skovoroda, A., Emelianov, S., Fowlkes, J., Pipe, J., Adler, R., Buxton, R., and Carson, P. (1995) Biophysical bases of elasticity imaging. In: *Acoustical imaging*. Springer, pp. 223–240.
- Sarvazyan, A., Hall, T.J., Urban, M.W., Fatemi, M., Aglyamov, S.R. & Garra, B.S. (2011) An overview of elastography – an emerging branch of medical imaging. *Curr Med Imaging Rev.* 7: 255–282.
- Sarvazyan, A.P., Rudenko, O.V., Swanson, S.D., Fowlkes, J.B. & Emelianov, S.Y. (1998) Shear wave elasticity imaging: a new ultrasonic technology of medical diagnostics. *Ultrasound Med Biol.* 24: 1419–1435.
- Shah, J.P., Danoff, J.V., Desai, M.J., Parikh, S., Nakamura, L.Y., Phillips, T.M. & Gerber, L.H. (2008) Biochemicals associated with pain and inflammation are elevated in sites near to and remote from active myofascial trigger points. *Arch Phys Med Rehabil.* 89: 16–23.
- Schleip R, Duerselen L, Vleeming A, et al. (2012) Strain hardening of fascia: static stretching of dense fibrous connective tissues can induce a temporary stiffness increase accompanied by enhanced matrix hydration. *J Bodyw Mov Ther.* 16(1): 94–100. doi:10.1016/j.jbmt.2011.09.003
- Toumi, H. & T. M. Best (2003) The inflammatory response: friend or enemy for muscle injury? *Brit J Sports Med.* 37: 284–286.